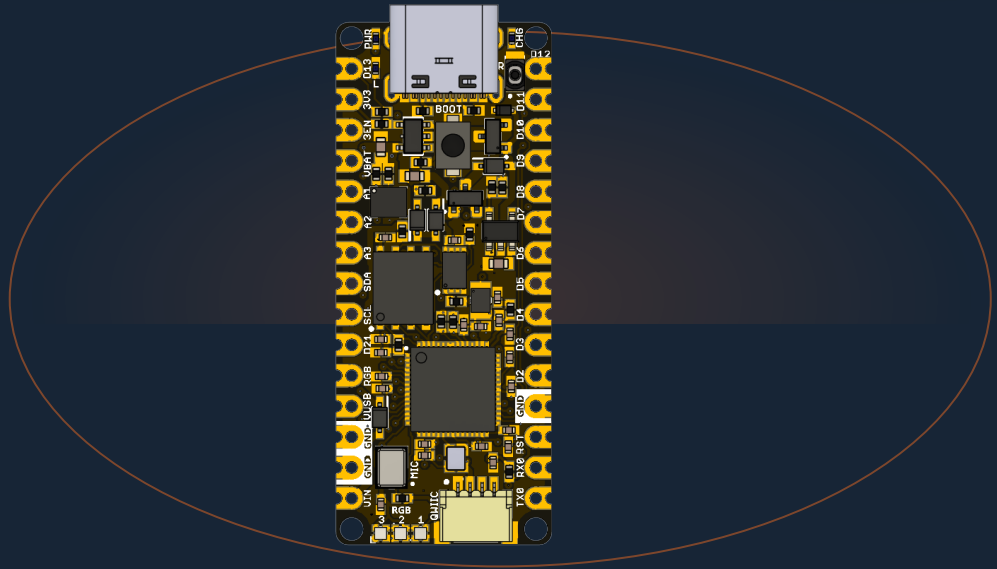


DEVLAB MULTI-INTERFACE DEVELOPMENT BOARD

UNIT DevLab PULSAR RP2350A

Mfr. Part #: UE0103

Version 0.1.0



Preview RP2350A development board with external flash and PSRAM, motion sensing, PDM audio, microSD, QWIC, USB-C, and HSTX expansion.

RP2350A

16 MiB FLASH

8 MiB PSRAM

SDIO · PDM · I2C · HSTX

Contents

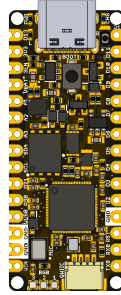
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Description

The UNIT DevLab PULSAR RP2350A is a multi-interface development board built around the Raspberry Pi RP2350A microcontroller. The pre-production V1.3 design combines external QSPI flash and PSRAM with motion sensing, PDM audio, microSD, USB-C, battery support, addressable RGB indicators, QWIIC I2C, and a 22-pin HSTX expansion connector.



Applications

- RP2350 firmware and peripheral prototyping
- Motion and PDM audio acquisition
- microSD data logging
- HSTX graphics experiments
- External-memory and multicore applications
- Education and subsystem validation

Hardware Features

- RP2350A microcontroller with external W25Q128 128 Mbit QSPI flash
- APS6404L-3SQR-ZR 8 MiB PSRAM
- BMI270 six-axis inertial measurement unit
- ICS-41350 digital PDM microphone
- Four-bit SDIO microSD connection
- Three WS2812-compatible RGB LEDs plus power, charge, and user indicators
- USB-C, QWIIC, battery, edge-pad, SWD, and HSTX connections

The hardware design is complete and awaiting fabrication. This Product Reference is a preview; values requiring measurement on manufactured units are marked pending confirmation.

1 The Board

1.1 Board Identification

Item	Value
Product	UNIT DevLab PULSAR RP2350A
Product type	Multi-interface development board
Manufacturer Part Number	UE0103
Main component	Raspberry Pi RP2350A
Hardware artwork	V1.3.0
Product Reference	0.1.0, preview

1.2 Confirmed Assemblies

RefDes	Fitted component	Board function
IC1	W25Q128JVP1Q	128 Mbit (16 MiB) QSPI flash
IC3	RP2350A	Main microcontroller
IC4	APS6404L-3SQR-ZR	8 MiB PSRAM
IC5	BMI270	Six-axis IMU
MK1	ICS-41350	Digital PDM microphone
U1	AP2112K-3.3TRG1	Fixed 3.3 V LDO
IC2	MCP73831T-2ACI/OT	Single-cell charge controller
LED1-LED3	WS2812 1010	Addressable RGB indicator chain
MICRO_SD-HOLDER	47309-2651	microSD socket

1.3 Package Contents

Package contents have not been formally released. No cable or battery is claimed as included in this preview edition.

2 Electrical Characteristics

2.1 Confirmed Component and Rail Values

Parameter	Value	Scope / evidence
Regulated board rail	3.3 V nominal	AP2112K-3.3TRG1 output shown in schematic and BOM
USB supply net	+5 V nominal	USB-C VBUS net label in schematic
External flash capacity	128 Mbit (16 MiB)	W25Q128JVPIQ part identification
External PSRAM capacity	8 MiB	APS6404L-3SQR-ZR schematic annotation and BOM
RP2350 internal SRAM	520 KiB	RP2350 component datasheet
RP2350 nominal maximum system clock	150 MHz	RP2350 component datasheet; not a module test result
Board oscillator	12 MHz	XTAL1 schematic and BOM
PDM microphone supply range	1.65 to 3.63 V	ICS-41350 BOM description; component rating only

The 3.3 V rail, USB net label, microphone supply range, and individual power component ratings do not establish complete board ratings.

2.2 Pending Module-Level Characteristics

- Allowed VIN and VBAT operating and absolute-maximum ranges
- 3.3 V rail output current available to external loads
- USB, battery, and VIN source-selection behavior under all conditions
- Battery charge current and compatible cell/cable specification
- GPIO thresholds, drive limits, and mixed-supply behavior at board connectors
- Total current consumption and thermal limits
- Validated SDIO, HSTX, PDM, PSRAM, and QSPI clock rates

2.3 Electrical Precautions

- Do not infer a board input limit from the AP2112K or MCP73831 component rating.
- Do not attach a battery until polarity, chemistry, connector fit, and charge settings have been validated for the assembly.
- Connect a common ground before applying external signals.
- Treat all exposed I/O as belonging to the 3.3 V RP2350A domain unless a released board specification states otherwise; tolerance to other levels is pending validation.

3 Functional Overview

3.1 Processing and Boot Memory

The RP2350A is the central controller for the board and provides the GPIO, ADC, I2C, PIO, USB, and HSTX resources used by the onboard subsystems. Program storage is provided by the external W25Q128JVPIQ QSPI flash, with a capacity of 128 Mbit (16 MiB). The BOOT pushbutton controls the flash boot-selection path, and the reset control restarts the RP2350A without removing board power.

The board is intended to be programmed through USB-C using the UNIT RP2040/RP2350 board package or another RP2350-compatible toolchain. The wiki uses the user LED as the first upload and execution check before enabling the remaining peripherals.

3.2 Internal and External Memory

The RP2350A provides 520 KiB of internal SRAM for low-latency program data. An APS6404L-3SQR-ZR adds 8 MiB of external PSRAM for framebuffers, acquisition buffers, large arrays, and other data that would otherwise consume internal SRAM. Its chip-select is connected to GPIO0.

In the Arduino workflow described by the wiki, selecting PSRAM CS = GPIO 0 enables the board definition to initialize this memory. Applications can then place static buffers in PSRAM or request dynamic allocations from its heap. Time-critical data can remain in internal SRAM while display buffers, logged data, and other large objects are kept in PSRAM.

3.3 Motion Sensing and I2C Expansion

The onboard BMI270 six-axis IMU uses the internal I2C pair on GPIO8 (SDA) and GPIO9 (SCL). Firmware reads its accelerometer and gyroscope data over this bus for motion, orientation, and user-interface applications.

The QWIIC connector provides a separate I2C path on GPIO24 (SDA) and GPIO25 (SCL), together with 3.3 V and GND. This allows external sensors, EEPROMs, and other I2C peripherals to operate without taking over the BMI270 bus. The wiki documents scanning and device access on both available I2C pin pairs.

3.4 microSD Storage

The microSD socket is electrically connected to GPIO2–GPIO7 as CLK, CMD, and DAT0–DAT3, allowing the RP2350A to use the complete four-bit SDIO signal group. The current wiki examples use the SPI-compatible subset:

microSD operation	GPIO	Schematic signal
Clock	2	SDIO_CLK
Command / MOSI	3	SDIO_CMD
Data / MISO	4	SDIO_DAT0
Chip select in SPI mode	7	SDIO_DAT3

This arrangement supports filesystem operations such as card initialization, file creation, reading, directory listing, and sensor-data logging. GPIO5 and GPIO6 provide DAT1 and DAT2 when software uses the full four-bit SDIO bus.

3.5 HSTX Video Output

The RP2350A HSTX peripheral can serialize display data into four pseudo-differential TMDS channel pairs for DVI-compatible video. The wiki defines the routed pair order as follows:

TMDS channel	Positive GPIO	Negative GPIO
Clock	14	15
Data 0	18	19
Data 1	16	17
Data 2	12	13

These GPIOs are available through the 22-pin HSTX connector. When video output is active, the eight pins are driven as a group and must be treated as reserved for that function. The wiki describes a 320×240 RGB565 framebuffer workflow, drawing primitives, text rendering, and partial-screen updates; PSRAM is available for large application buffers where supported by the display stack.

3.6 PDM Audio Input

The onboard ICS-41350 converts acoustic pressure into a one-bit PDM data stream. The RP2350A supplies the microphone clock on GPIO10 and receives data on GPIO11. A PDM-capable software peripheral decimates this stream into signed PCM samples that can be processed, displayed, or stored on microSD.

The microphone is already connected on the PCB; no external audio wiring is required for onboard capture. Sample rate, buffering, filtering, and storage format are selected by the application software.

3.7 Visual Indicators and User Feedback

Three cascaded WS2812-compatible RGB LEDs share a single data input on GPIO1. Because the LEDs are connected as a serial chain, firmware sends one ordered color frame to update all three devices. They can provide status, progress, alerts, or application-specific color feedback.

The board also includes a GPIO20 user indicator (D13 / BUILTIN1), a power indicator, and a charge-status indicator. The RGB chain and user LED are independently controlled.

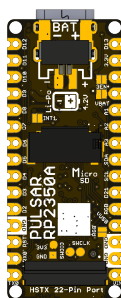
3.8 Power Distribution and Battery Support

USB VBUS, VIN, and the battery circuit feed the board power path. The AP2112K-3.3TRG1 generates the 3.3 V rail used by the RP2350A and onboard peripherals. The MCP73831 provides the single-cell battery-charging function, with the charge indicator reporting the controller status. Schottky diodes and MOSFETs implement the source-routing and protection paths shown in the schematic.

The hardware design is complete and awaiting fabrication. Final module-level input ranges, available rail current, charge configuration, and source-priority behavior will be confirmed on the first manufactured units.

3.9 Combined System Operation

The subsystems are intended to operate together: the BMI270 or PDM microphone can provide acquisition data, PSRAM can hold large working buffers, microSD can store results, the RGB LEDs can report state, and HSTX can present a graphical interface. The wiki's complete-application chapter combines motion sensing and video output and describes moving large buffers out of internal SRAM to keep the RP2350A responsive.



4 Connectors and Pinout

4.1 Edge Connections

Board label	RP2350A connection	Function / status
TX0 / D1	GPIO18	Digital I/O / UART-capable
RX0 / D0	GPIO19	Digital I/O / UART-capable
D2, D3, D4, D5, D6, D7	GPIO17, 16, 15, 14, 13, 12	Digital I/O; also routed to HSTX connector
D8, D9	Pending	Artwork labels exist; schematic shows no RP2350A connection
D10 / SS	GPIO21	Digital I/O
D11 / MOSI	GPIO22	Digital I/O
D12 / MISO	GPIO23	Digital I/O
D13 / SCK / LED	GPIO20	Digital I/O and user indicator
A1 / D15	GPIO29 / ADC3	Analog-capable digital I/O
A2 / D16	GPIO27 / ADC1	Analog-capable digital I/O
A3 / D17	GPIO26 / ADC0	Analog-capable digital I/O
SDA / D18	GPIO8	Internal/user I2C data
SCL / D19	GPIO9	Internal/user I2C clock
D21	GPIO10	PDM clock net
RGB	GPIO1	WS2812 data
3V3, 3EN, VBAT, VUSB, VIN	Power	See Section 2; operating limits pending
RST	Reset	RP2350A reset control
GND	Ground	Common return

4.2 Fixed Peripheral Mapping

Peripheral	Signal mapping
PSRAM	CS=GPIO0; data/clock on the schematic QSPI nets
microSD	CLK=GPIO2, CMD=GPIO3, DAT0=GPIO4, DAT1=GPIO5, DAT2=GPIO6, DAT3=GPIO7
BMI270	SDA=GPIO8, SCL=GPIO9
PDM microphone	CLK=GPIO10, DATA=GPIO11
HSTX signal group	GPIO12 through GPIO19
QWIIC	SDA=GPIO24, SCL=GPIO25, 3.3 V, GND
RGB LED chain	DATA=GPIO1

4.3 QWIIC Connector

J1 is a four-position, 1 mm-pitch right-angle connector. The schematic shows GND, 3.3 V, GPIO24/SDA, and GPIO25/SCL. A controlled contact-number/orientation drawing and compatible cable specification are pending validation.

4.4 HSTX Connector

J5 is a 22-position, 0.5 mm-pitch FFC/FPC connector. It exposes D0–D7, A0/GPIO28, A1/GPIO29, GPIO24/SDA, GPIO25/SCL, 3.3 V, and return positions. Use the released schematic for prototype tracing; the complete controlled connector pin-number table remains pending validation.

4.5 Programming and Battery Connections

The bottom artwork shows SWDIO, SWCLK, 3.3 V, and GND programming pads. JP1 is a two-position battery connection. Battery polarity is marked on the artwork, but a compatible battery/cable assembly is not specified by this preview.

5 Pre-production Operation Notes

The technical wiki documents the intended firmware workflow and subsystem operation. Those examples are not reproduced in this Product Reference; their results will be confirmed on the first manufactured units. The complete sketches are maintained under `software/cpp_examples/`.

For initial bench work:

1. Inspect the board and confirm its artwork revision.
2. Review the V1.3 schematic and the pending electrical limits in Section 2.
3. Power through USB-C for first inspection; do not connect a battery or an external VIN source until its limits and polarity have been validated.
4. Establish a common ground before attaching test equipment or peripherals.
5. Validate the user LED and serial upload path before enabling external memory, storage, sensors, audio, or HSTX output.
6. Confirm each peripheral mapping against Section 4 and observe the signals with suitable test equipment before relying on captured data.

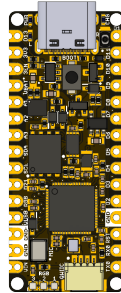
Host framework aliases, GPIO initialization, PSRAM setup, SDIO support, and HSTX library behavior depend on the selected board package. The wiki is an implementation work-in-progress and may contain mappings that predate the V1.3 schematic review.

5.1 Current Wiki Coverage

- Board setup and LED validation
- ADC and WS2812 examples
- Internal and external I2C examples
- microSD experiments
- HSTX display experiments
- PSRAM allocation examples
- PDM microphone capture
- A combined demonstration application

Results, performance claims, and troubleshooting instructions from these chapters remain pending confirmation on manufactured hardware.

6 Mechanical Information



The artwork shows two parallel edge-pad rows, four corner mounting holes, USB-C at one end, and the HSTX connector at the opposite end. The microSD and battery connectors are mounted on the bottom side.

A controlled dimension drawing, board thickness, keep-out definitions, connector envelopes, and mounting-hole coordinates were not supplied. All mechanical dimensions are pending validation; do not scale them from the rendered images.

7 Company Information

Item	Information
Company	UNIT Electronics
Website	UNIT Electronics
Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX, Mexico

Product specifications and documentation may change during preview. Consult the repository and Product Wiki in the next chapter for current released sources.

8 Reference Documentation

Reference	Location
Product repository	UNIT-Electronics-MX/unit_pulsar_rp2350a
Hardware reference	hardware/README.md
Technical wiki	PULSAR RP2350 wiki
Getting Started	Preview setup guide
C++ examples	software/cpp_examples
Board schematic	V1.3 schematic
RP2350 component datasheet	Raspberry Pi RP2350 Datasheet

The wiki is the implementation guide for the board's intended operation, but it is not a source of module-level electrical ratings. The supplied legacy Product Reference DOCX is also excluded as a technical source because its content describes a PULSAR ESP32-H2 rather than this RP2350A board.

9 Appendix

9.1 Source Hierarchy

1. V1.3 schematic, manufacturing BOM, and V1.3 board artwork for fitted components, visible labels, and board connections.
2. Official Raspberry Pi RP2350 datasheet for RP2350A component characteristics only.
3. Technical wiki for intended subsystem operation and firmware workflow.

9.2 Items Pending Validation

- Correction of the schematic title block from PULSAR RP230A / REV 1.0.0 to match the RP2350A V1.3 source package
- Module VIN, VBAT, USB, and 3.3 V rail electrical limits
- Power-source priority, battery charge current, and compatible battery cable
- Complete controlled HSTX and QWIIC connector pin-number/orientation drawings
- D8 and D9 edge-pad connectivity
- Board dimensions, mounting-hole coordinates, and mechanical keep-outs
- Validated clock rates, current consumption, and thermal behavior
- Reconciliation and test of all technical-wiki examples against V1.3 hardware
- Controlled pinout, topology, and dimensions artwork

9.3 Document Control

Field	Value
Product	UNIT DevLab PULSAR RP2350A
Manufacturer Part Number	UE0103
Hardware artwork covered	V1.3.0
Product Reference	0.1.0, preview
Publication date	2026-08-03
Hardware status	Design complete; awaiting fabrication
Firmware status	Implementation documented in the wiki; results pending confirmation on manufactured units

9.4 Source Inconsistencies

- The V1.3 schematic filename, artwork, and BOM identify RP2350A, but the schematic title block says PULSAR RP230A and revision 1.0.0.
- The supplied Product Reference DOCX describes an ESP32-H2 board and is not applicable to PULSAR RP2350A.
- Some wiki chapters describe SPI-oriented microSD examples, while the V1.3 schematic shows a four-bit SDIO connection on GPIO2–GPIO7. This edition follows the schematic and leaves software behavior pending validation.